Domain-Trained Enterprise RAG

An Evaluation & Intelligence Platform



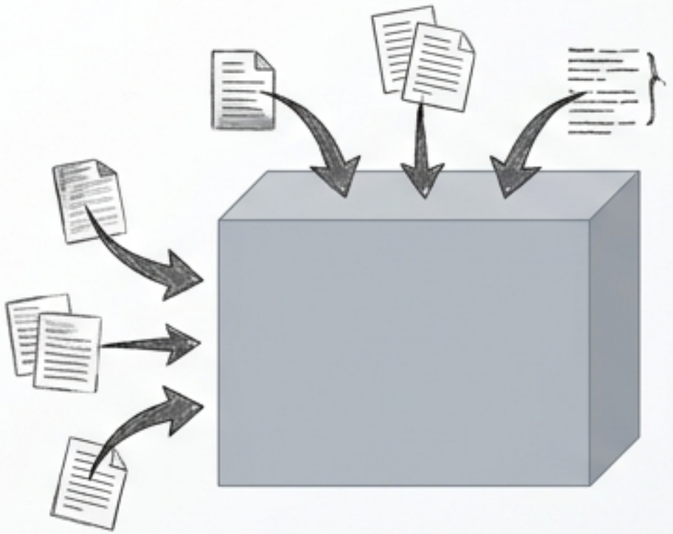
Owner
Mazhar Ali

Scope
Advanced 8-Week Capstone

Focus
Retrieval Fine-Tuning, Reranking, LLM Fine-Tuning & RAG Security

Gemini Notebook

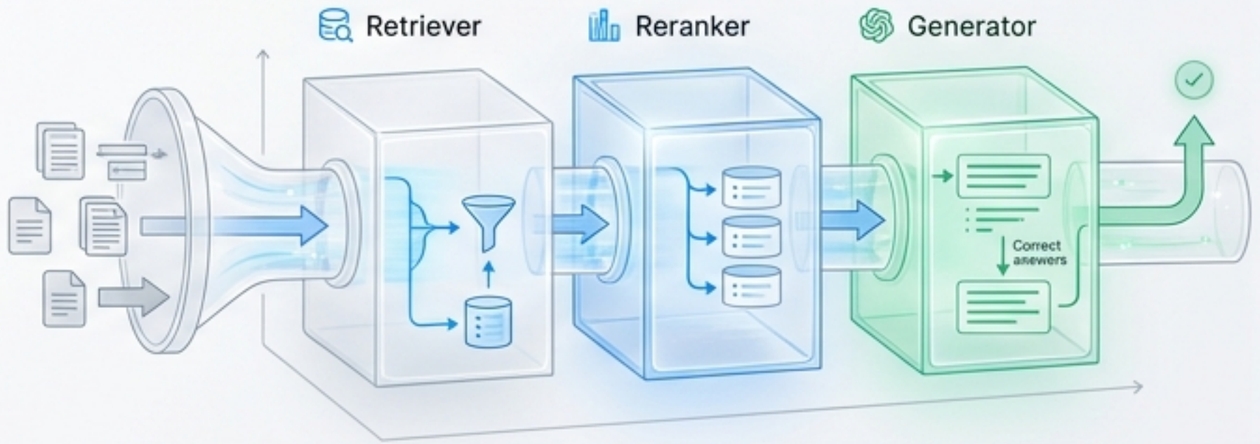
Assembled vs. Trained Architecture



Naive RAG

Merely assembled. Unmeasured, vulnerable to noise, and blind to domain nuances.

The Intelligence Platform



Systematically trained and evaluated. Every component—Retriever, Reranker, and Generator—is optimized and measured.

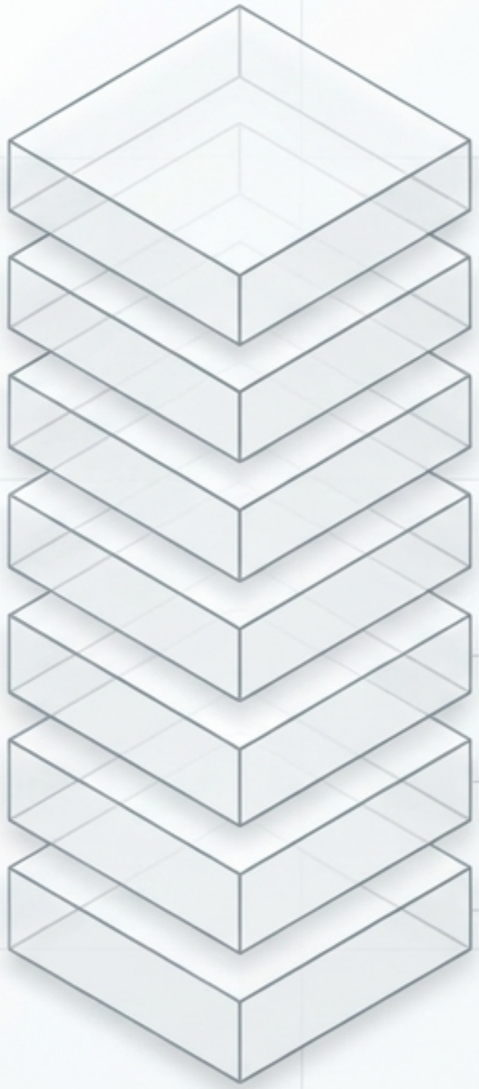
The Primary Research Question

Where does domain fine-tuning create measurable value in an enterprise RAG stack, and how robust is the resulting system to misleading, outdated, and adversarial knowledge?

Gemini Notebook

The 7 Progressive Layers of Intelligence

The user sees a single application. Internally, the architecture executes seven distinct stages of enhancement.



- L1: Base LLM
- L2: Naive RAG
- L3: Fine-Tuned LLM
- L4: Advanced RAG
- L5: FT LLM + Advanced RAG
- L6: Trained Retriever + Reranker
- L7: Production Defense Layer

Key Insight

The project is not seven independent demos. It is a single, evolving pipeline where every layer's exact ROI is mathematically proven against a frozen benchmark.

Gemini Notebook

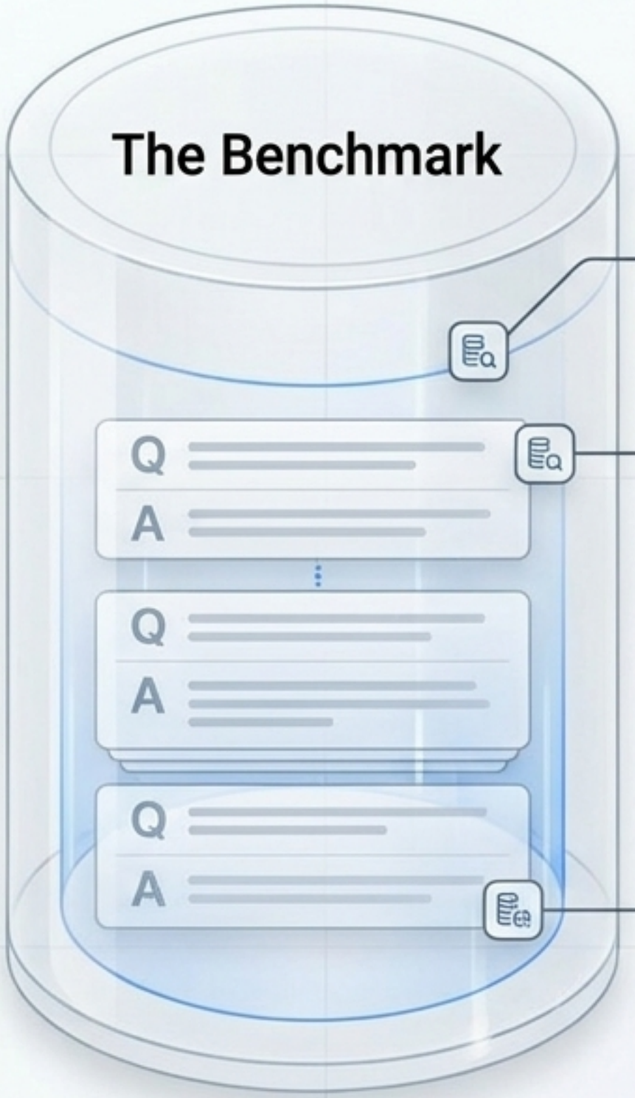
Foundation: The Corpus and The Benchmark



10,000+ meaningful sections/chunks (Cleaned & Deduplicated)

Spans DevOps/Cloud/Infra: Linux, Nginx, Docker, Kubernetes, PostgreSQL, TLS, CI/CD





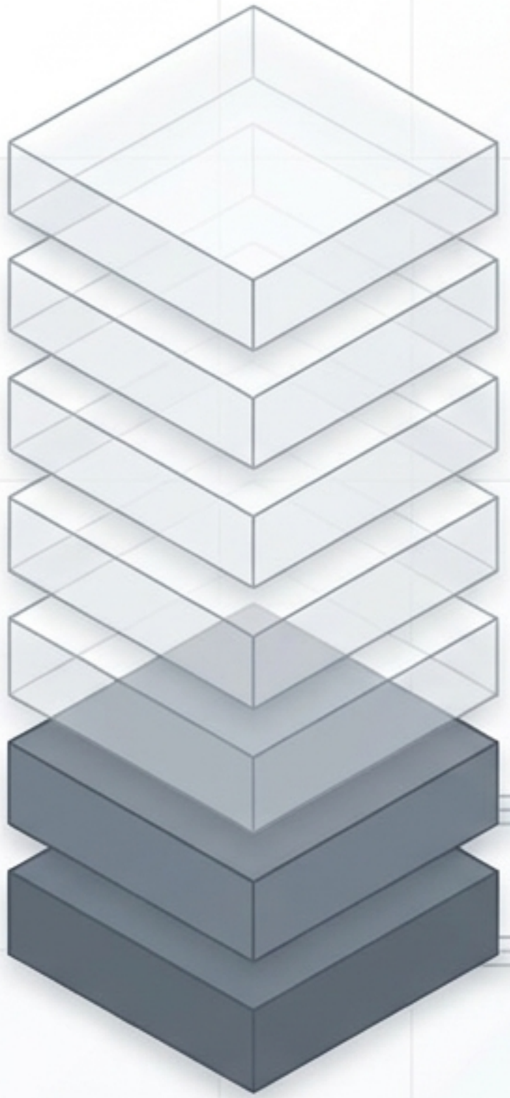
1,000+ categorized questions designed before tuning

Categories: Direct, Multi-hop, Troubleshooting, Cross-document, Version-sensitive, Ambiguous, Unanswerable, Distractor-heavy

Includes visible development split & mentor-held hidden set

Gemini Notebook

Establishing the Baseline



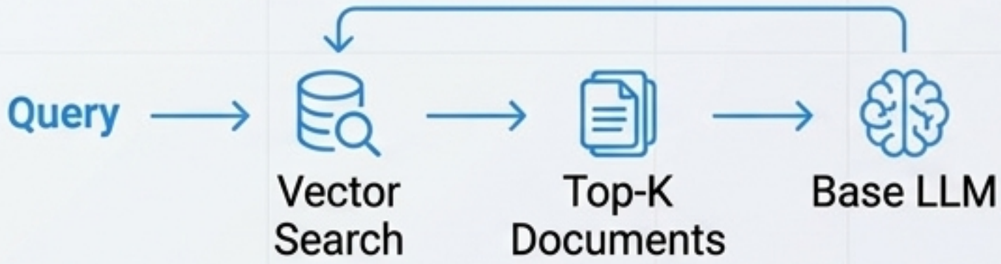
L2: Naive RAG

L1: Base LLM

Layer 1: Base LLM

No retrieval, no fine-tuning. Establishes the absolute baseline for latency and hallucination on pretrained knowledge.

Layer 2: Naive RAG



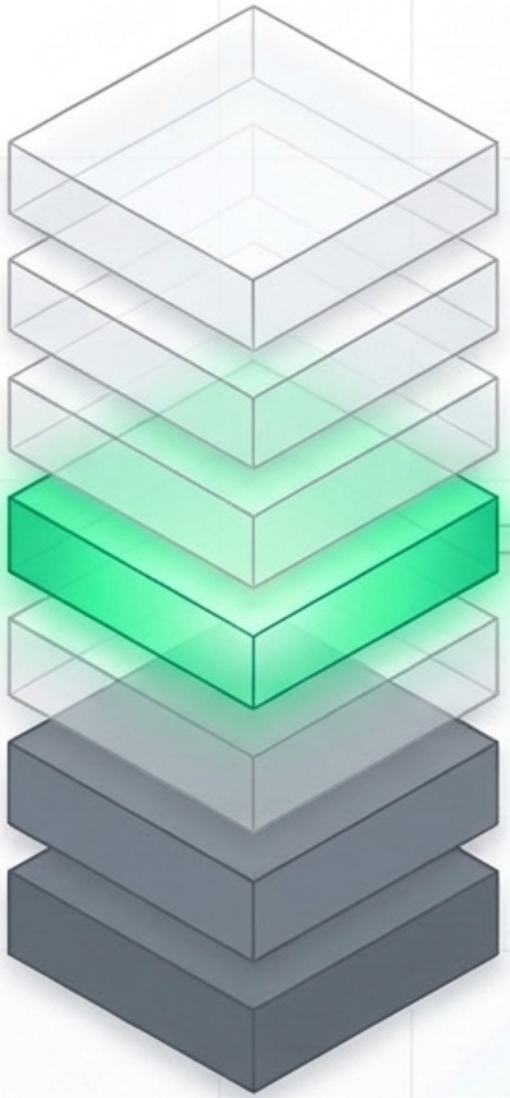
Introduces retrieval. Implements BM25, generic dense embeddings, and a hybrid baseline.

The Vulnerability

While it grounds answers, it blindly trusts generic embeddings and struggles with domain-specific jargon.

Gemini Notebook

Layer 3: Domain Generator Fine-Tuning



Generic Context

QLoRA / SFT

Low-Rank Adaptation / Supervised Fine-Tuning

Highly Structured Response

Strategy

Keep the retrieval controlled; upgrade the generator.

Training Data Composition

Several thousand validated examples containing:

Question + Context + Correct Answer + Citations.

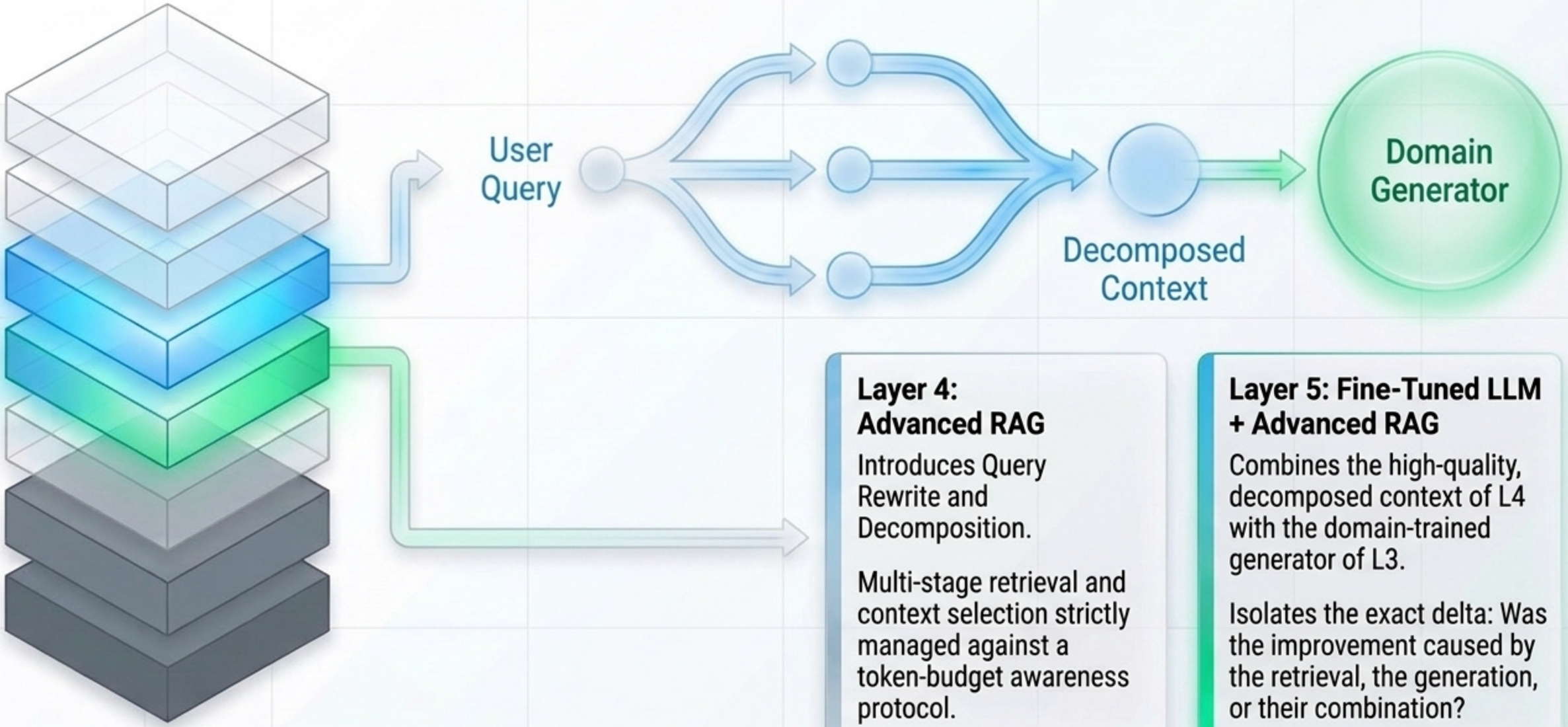
Behavioral Training & Result

Explicitly trained on Refusal Examples (for unanswerable queries) and Multi-hop Reasoning.

The generator natively understands DevOps context and correctly cites evidence or fails gracefully.

Gemini Notebook

Layers 4 & 5: Advanced Retrieval & Synergy



Layer 4: Advanced RAG

Introduces Query Rewrite and Decomposition.

Multi-stage retrieval and context selection strictly managed against a token-budget awareness protocol.

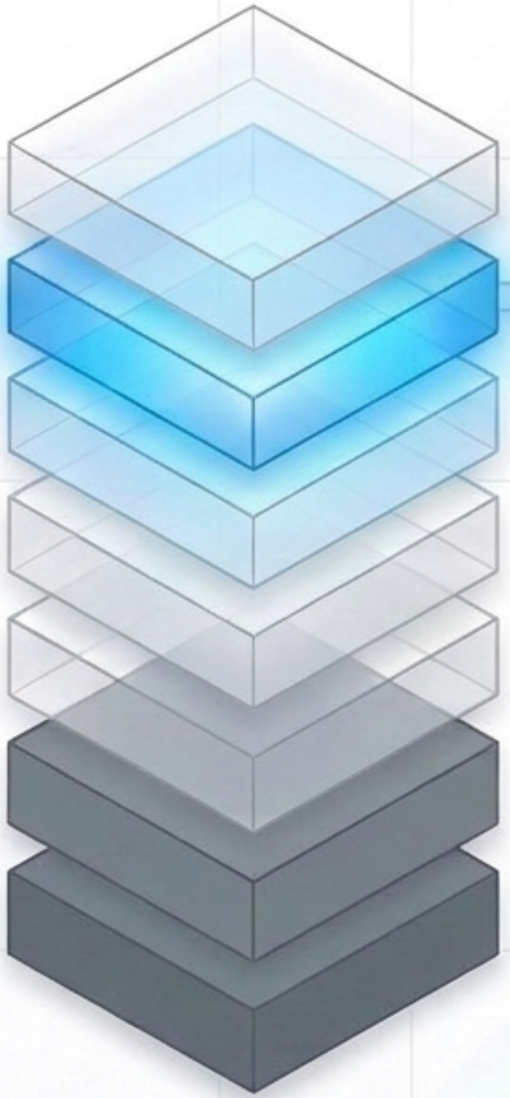
Layer 5: Fine-Tuned LLM + Advanced RAG

Combines the high-quality, decomposed context of L4 with the domain-trained generator of L3.

Isolates the exact delta: Was the improvement caused by the retrieval, the generation, or their combination?

Gemini Notebook

Layer 6: Training the Retriever and Reranker



Query: Why is my Kubernetes pod repeatedly restarting after deployment?

Positive

The exact troubleshooting guide for CrashLoopBackOff.

Soft Negative

Generic Kubernetes overview (easily ignored).

Hard Negative

A document about restarting a Docker container (semantically similar, extremely tricky, but wrong).

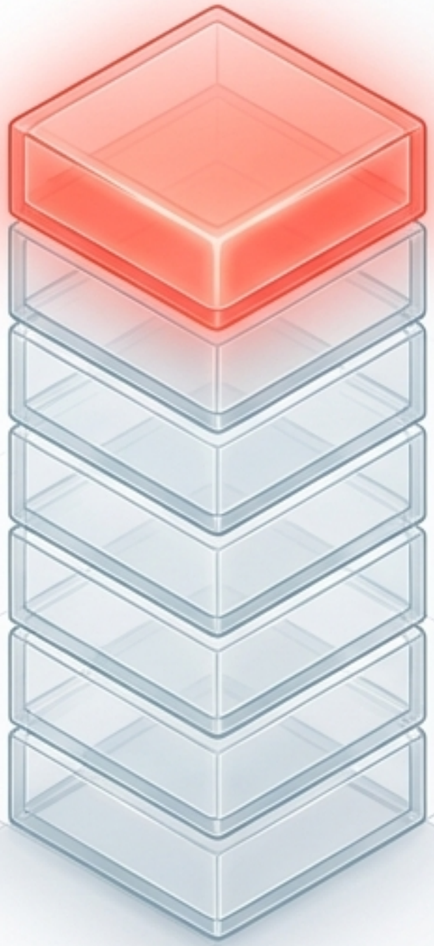
Bi-Encoder Retriever: Fine-tuned using Query-Positive-Hard-Negative triples over at least two mining iterations.

Cross-Encoder Reranker: Domain-adapted to compare generic vs. domain reranking top-k trade-offs.

The Outcome: The system learns to ignore near-identical but technically incorrect documents.

Gemini Notebook

Layer 7: The Production Defense Shield



The Trust Classifier

Treats input as raw data, not instructions. Explicitly separates instruction from data to prevent prompt injection.



Claim-Level Citation Verifier

Classifies every generated claim as SUPPORTED, PARTIAL, or UNSUPPORTED.

Fail-Closed Logic

Automatically regenerates or refuses to answer if claims are unsupported.

Gemini Notebook

Adversarial Survival: The Security Suite

Attack Vector	Automated Defense Mechanism
Prompt Injection (e.g., “Ignore previous instructions...”)	Filtered / Handled strictly as raw text
Contradictory Documents	Flagged / Down-ranked by verification logic
Outdated Knowledge	Deprioritized based on version-sensitive metadata
Poisoned / Irrelevant Documents	Caught by domain-adapted cross-encoder reranker
Near-Duplicates	Cleaned during ingestion and context selection

Result: Operation within a fixed local inference budget with heavily bounded hallucination risk.

Gemini Notebook

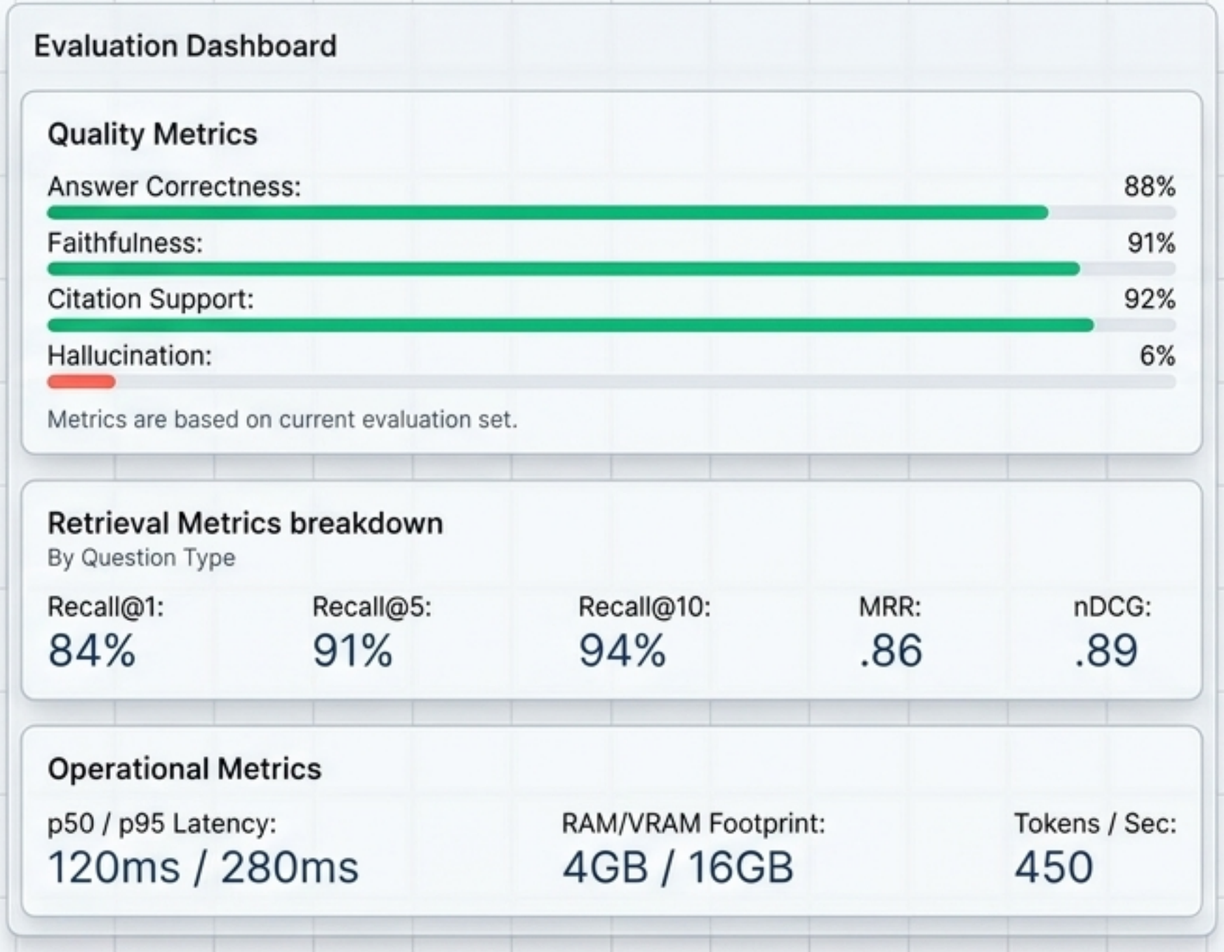
The Glass Box Laboratory

The Live Comparison Matrix

Metric	L1 Base	L2 Naive	L3 FT	L4 Adv	L5 Combined	L6 Full	L7 Final
Recall@5	—	61%	—	74%	74%	89%	89%
MRR	—	.48	—	.61	.61	.78	.78
Correctness	54%	68%	71%	77%	83%	88%	90%
Faithfulness	42%	71%	67%	82%	88%	91%	95%
Citation Support	**0%**	**66%**	**0%**	**81%**	**87%**	**92%**	**97%**
Hallucination	**31%**	**18%**	**16%**	**12%**	**8%**	**6%**	**3%**

Gemini Notebook

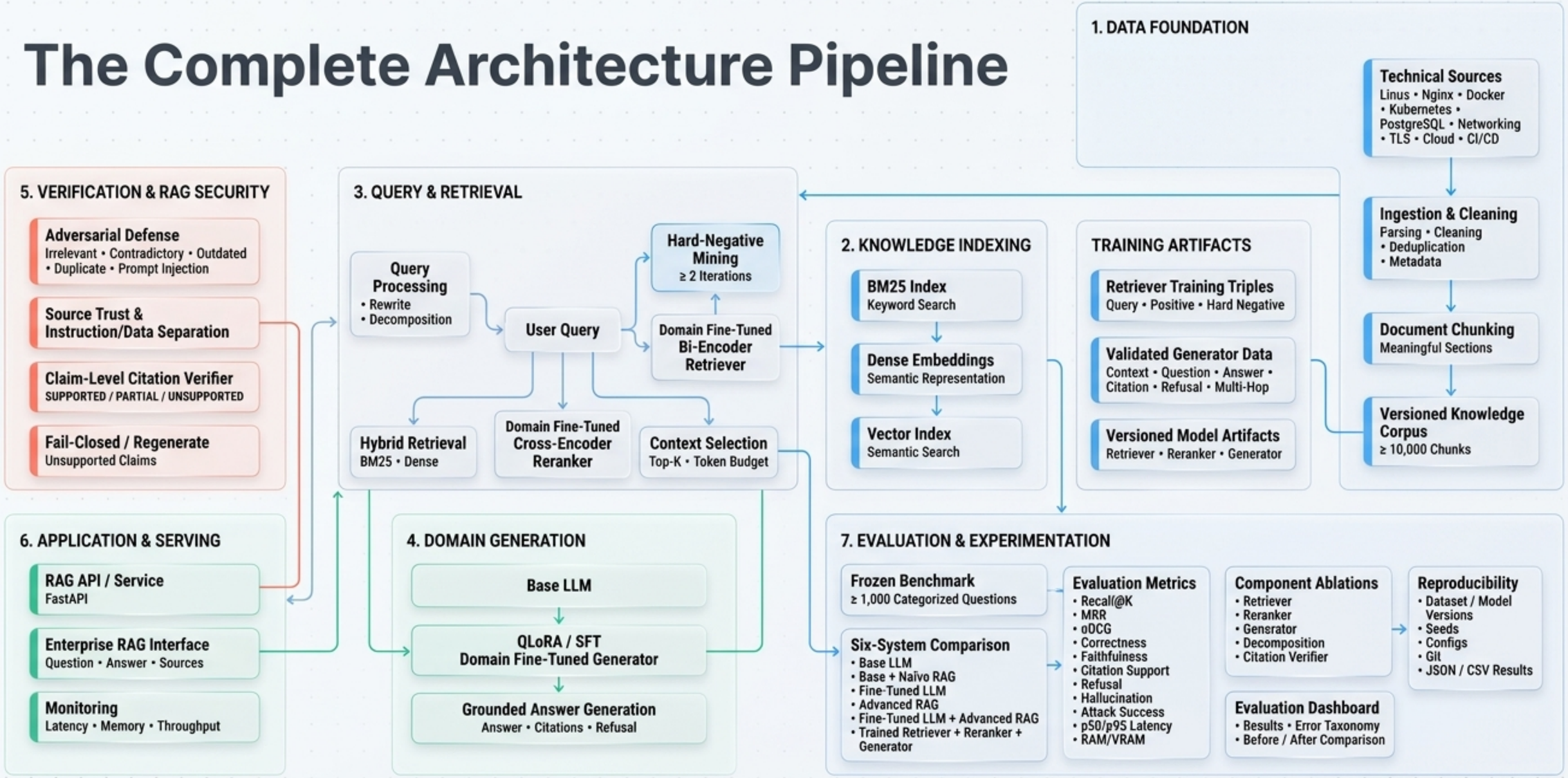
The Evaluation Engine



Key Insight: A repeatable evaluation command produces machine-readable JSON/CSV results and human-readable charts, completely separating the benchmark from normal user queries.

Gemini Notebook

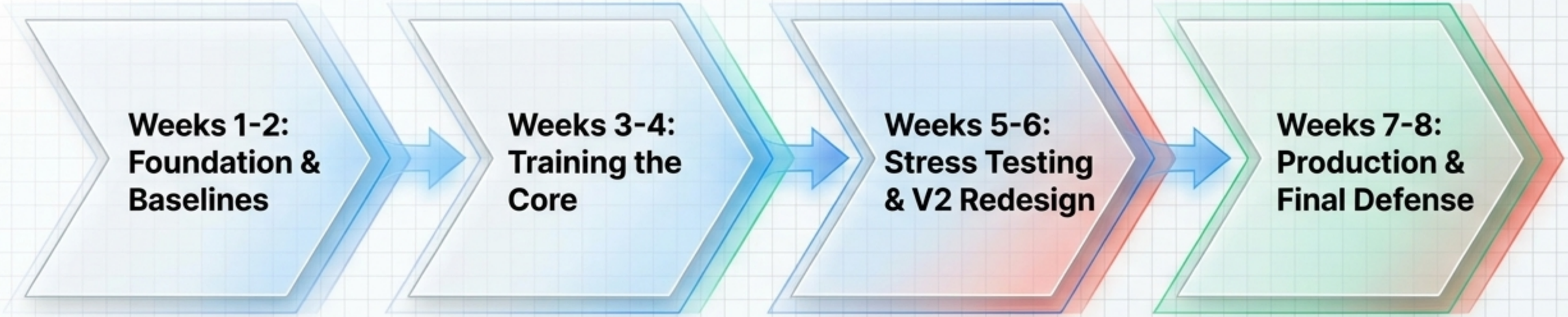
The Complete Architecture Pipeline



- Traceability from raw Technical Sources to the Final API Service.
- Continuous parallel evaluation alongside the main user request path.

Gemini Notebook

The 8-Week Execution Roadmap



Corpus frozen, BM25/Dense baselines evaluated.

Retriever/Reranker training, Hard Negative Mining V1/V2, QLoRA SFT.

Structured failure discovery, measurable ablations without regression.

Containerization, hidden benchmark, viva.

Gemini Notebook

The Ultimate Standard: The Hidden Benchmark



Dev Set

Hidden Set

Code Freeze

No prompt hand-tuning. The system must generalize.

Unseen Challenge

Mentor-injected private questions and hostile, misleading documents post-freeze.

Deliverable Complete

Containerized local inference package, 15-20 page technical report (with error taxonomy), and a live technical defense of the ablation results.

Building an Enterprise RAG platform is not about the first working demo—it is about relentless measurement, failure discovery, and architectural iteration.

Gemini Notebook